EC-350 AI and Decision Support Systems

Week 4
Solving Problems by Searching

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Iterative Deepening DFS

- General search strategy used with DFS
- We look for the solution at first level and then gradually increase the depth limit
- First 0, then 1, then 2 and so on
- Combines the benefits of BFS and DFS

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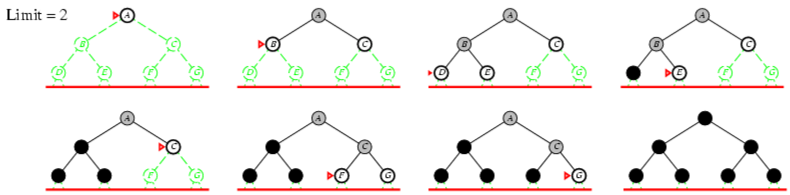
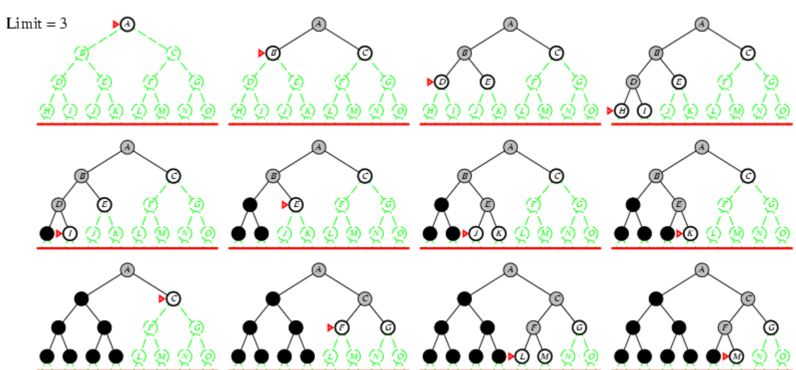
Iterative Deepening Search *l* =0

Iterative Deepening Search *l* =1

Iterative Deepening Search l =2



Iterative Deepening Search l =3



Properties of Iterative Deepening Search

- Complete? Yes
- Time? $(d+1)b^0 + d\,b^1 + (d-1)b^2 + \ldots + b^d = O(b^d)$
- Space? $O(bd)$
- Optimal? Yes, if step cost = 1

Iterative Deepening Search

- No of nodes generated in a Breadth-first search to depth d with branching factor b:

  $N_{BFS} = b^0 + b^1 + b^2 + \ldots + b^d$

- No of nodes generated in an iterative deepening search:

  $N_{IDS} = (d+1)b^0 + db^1 + (d-1)b^2 + \ldots + b^d$

- For $b = 10$, $d = 5$
  - $N_{BFS} = 1+10 + 100 + 1{,}000 + 10{,}000 + 100{,}000 = 111{,}111$
  - $N_{IDS} = 6+50 + 400 + 3{,}000 + 20{,}000 + 100{,}000 = 123{,}456$
- In general IDS is the preferred uninformed search method when there is a large search space and the depth of the solution is not known

Example

- Find path from S to E
- Using IDS